Plane scoring

How AirWSJT-X decides which aircraft are candidates, how they are ranked, and when auto-lock and tracking act on the score. Source: `airwsjtx/geo.py` and `sidecar/airwsjtx_bridge.py`.

The score

The candidate list, auto-lock and the percentage shown in the UI all use `scatter_geometry()`. It returns a score from 0 to 1, displayed as 0–100%.

Hard gate: if the plane is at or below the horizon from station A **or** station B (elevation ≤ 0°), the score is 0.

Otherwise the score is a weighted sum of four parts:

Weight	Part	Computation
30%	Elevation from A	`el / 20` up to 20°, then falls linearly to 0 at 40°. Best at 20°.
30%	Elevation from B	Same curve, using B's elevation.
25%	Path alignment	`1 − off_path / 45°`, where `off_path` is the angle between the A→plane bearing and the A→B bearing. 0 at 45° or more off.
15%	Range	`1 − max(slant range A, slant range B) / 800 km`. Closer is better.

Filters

`_aircraft_matches_filters` decides whether a plane is shown or auto-locked at all. These apply in addition to the score:

- **Lat/lon box:** optional minimum and maximum values.
- **Minimum altitude:** optional, in feet.
- **Position tolerance:** default 70°. The plane's bearing from A must be within this of the A→B bearing. Skipped when A and B are the same point.
- **Heading tolerance:** default 30°. If the plane reports a heading and speed, its heading must be within this of the A→B bearing.

How the score is used

- **Auto-lock:** runs only from idle, completed or aborted states. Picks the highest score of at least 0.05 among planes that pass the filters.
- **Tracking:** every update recomputes the score. Below 0.05 the plane is out of window: `plane_out_of_window` is emitted and the plane is unlocked.
- **Sorting:** the candidate list is sorted by score, highest first.

Limitations

- **Station altitudes:** the bridge passes both MY (station A) and DX (station B) altitudes to `scatter_geometry`. Each defaults to 0 m unless entered in the Stations panel.
- **Altitude:** apart from the optional minimum-altitude filter, altitude only enters through the elevation angles.
- **Speed, heading, vertical rate:** these do not affect the score. Heading acts only as a pass/fail filter.
- **Separate ranking:** `airwsjtx/tracker.py` has a different ranking based on average elevation, range and track stability. Only the `rank_candidates` RPC and the supervisor use it. The UI and auto-lock do not.